Worksheet 4

Exercise 1: Shoe Size

The dataset `d.shoes` in `shoes.rda` contains gender, income and shoe size information for 40 individuals.

a) Fit a simple linear regression model to the data, with income (`salary`) as the response variable and shoe size (`shoes`) as the explanatory variable. How do you interpret the result?

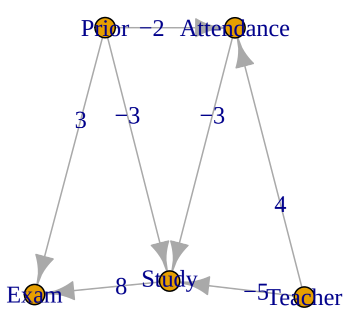
b) Fit a multiple linear regression model to the data, with income (`salary`) as the response variable and shoe size (`shoes`) as well as gender (`gender`) as the two explanatory variables. Compare the result with subtask a).

c) Use the regression models from a) and b) to calculate the expected income for a woman with a shoe size of 38. Explain the difference in the prediction.

Exercise 2: Exam Score

In this exercise, we will analyze the effect of study hours on the exam grade (continuation from Worksheet 2). We will introduce a bit more complexity by adding a node for teacher quality and student attendance in class. The following graphical causal model shows all dependencies:

```r
set.seed(253)
library(gRbase)
library(igraph)
grest <- dag(c("Study","Teacher","Prior","Attendance"),
             c("Attendance","Prior","Teacher"),
             c("Exam","Study","Prior"))
E(grest)$weight <- c(-5,-3,-3, -2, 4, 8, 3)
plot(grest, edge.label = E(grest)$weight)
```



Assume that the data generating process is given by linear structural causal model and the numbers in the DAG correspond to the coefficients.

a) What is the direct effect from studying hours on the exam score?

b) What is the total causal effect of class attendance on the exam score?

c) What is the total causal effect of prior knowledge on the exam score?

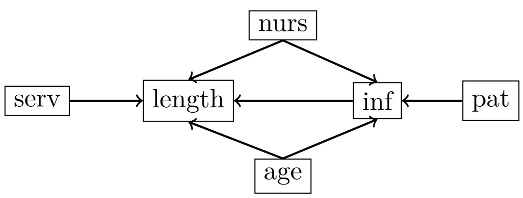
d) Report the R code to estimate the total causal effect of attendance on the exam score.

Exercise 3: Hospital

In a study on the risk of infection in hospitals, the following variables were recorded in a random sample of 112 U.S. hospitals:

`length`: average length of hospitalization of all patients (in days)
`age`: average age of the patients
`inf`: average risk of infection (in percent)
`pat`: average number of patients in hospital per day
`nurs`: proportion of fully employed and educated nurses per patient
`serv`: proportion of services offered per patient

The following graphical model describes the causal relations between these variables:



a) Fit a simple linear regression with the response `length` and the explanatory variable `inf`. What does the coefficient of `inf` tell us about the average length of hospitalization?

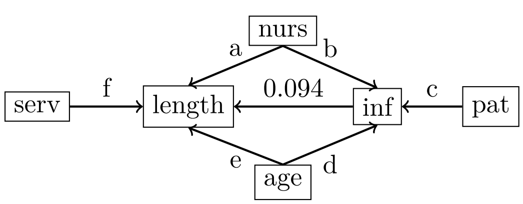
b) Why does the model in a) not measure the correct total causal effect of inf on length?

c) Determine at least two possible adjustment sets to estimate the true total causal effect of `inf` on `length`?

d) Estimate the total causal effect using at least two different adjustment sets. Do you see differences in the results?

e) Which adjustment set is optimal according to theory?

f) Calculate all causal effects (a, b, c, d, e and f) using appropriate regression models.



Exercise 4: Distributions (Optional)

Suppose the three variables X, Y and Z are described by the following SCM:
$X \leftarrow E_X$
$Y \leftarrow 1 + 2 \cdot X + E_Y$
$Z \leftarrow -1 + 3 \cdot X + 2 \cdot Y + E_Z$
$W \leftarrow 3 - 2 \cdot X + E_W$

$V \leftarrow 3 - 1 \cdot Y + 4 \cdot W + E_V$
where $E_X, E_Y, E_Z, E_W, E_V \sim \mathcal{N}(0, 1)$

a) Draw a causal diagram for the given SCM.

b) Simulate N = 10000 data points and visualize the distribution of V.

c) Adapt the simulation of b) to compute the interventional distribution of (V | do(X = 2)) and (V | do(X = 3)).

d) Determine the total causal effect of X on V by (1) calculating from the given SCM and (2) estimating from the simulated data (in c)).

e) What is the connection between c) and d)?

f) Which of the following statements is correct?

- The marginal distribution of $Y \sim \mathcal{N}(1, 5)$
- The interventional distribution of Y | do(X = 2) is equal to the marginal distribution of Y.
- The interventional distribution of X | do(Y = 1) is equal to the marginal distribution of X.
- The interventional distribution of Y | do(X = 2) is equal to the conditional distribution of Y | X = 2.